

Gamma and Zeta Functions

*Note: p is used exclusively for **prime numbers**.*

Theorem 1. *As $x \rightarrow \infty$, we have:*

$$\sum_{n \leq x} \omega(n) = x \log \log x + C_1 x + O\left(\frac{x}{\log x}\right)$$

and

$$\sum_{n \leq x} \Omega(n) = x \log \log x + C_2 x + O\left(\frac{x}{\log x}\right)$$

where $C_1 = \lim_{x \rightarrow \infty} \left(\sum_{p \leq x} \frac{1}{p} - \log \log x \right)$ is the Meissel-Mertens constant, and $C_2 = C_1 + \sum_p \frac{1}{p(p-1)}$.

Proof. First, since $\omega(n) = \sum_{p|n} 1$, we can interchange the sums:

$$\sum_{n \leq x} \omega(n) = \sum_{n \leq x} \sum_{p|n} 1 = \sum_{p \leq x} \left\lfloor \frac{x}{p} \right\rfloor$$

Using the fact that $\lfloor y \rfloor = y + O(1)$, we get:

$$\sum_{p \leq x} \left\lfloor \frac{x}{p} \right\rfloor = x \sum_{p \leq x} \frac{1}{p} + O\left(\sum_{p \leq x} 1\right) = x \sum_{p \leq x} \frac{1}{p} + O(\pi(x))$$

Using the prime number theorem bound $\pi(x) = O\left(\frac{x}{\log x}\right)$ and Mertens' Second Theorem $\sum_{p \leq x} \frac{1}{p} = \log \log x + C_1 + O\left(\frac{1}{\log x}\right)$, we obtain:

$$\sum_{n \leq x} \omega(n) = x \left(\log \log x + C_1 + O\left(\frac{1}{\log x}\right) \right) + O\left(\frac{x}{\log x}\right)$$

$$\sum_{n \leq x} \omega(n) = x \log \log x + C_1 x + O\left(\frac{x}{\log x}\right)$$

On the other hand, since $\Omega(n) = \sum_{p^k|n} 1$, we have:

$$\sum_{n \leq x} \Omega(n) = \sum_{p^k \leq x} \left\lfloor \frac{x}{p^k} \right\rfloor = x \sum_{p^k \leq x} \frac{1}{p^k} + O\left(\sum_{p^k \leq x} 1\right)$$

Now, we split the sum over prime powers:

$$\sum_{p^k \leq x} \frac{1}{p^k} = \sum_{p \leq x} \frac{1}{p} + \sum_{p \leq x} \sum_{k \geq 2}^{\lfloor \log_p x \rfloor} \frac{1}{p^k}$$

We can extend the inner sum to infinity, absorbing the tail into the error term. Using Mertens' Second Theorem again:

$$\begin{aligned}\sum_{p^k \leq x} \frac{1}{p^k} &= \log \log x + C_1 + O\left(\frac{1}{\log x}\right) + \sum_p \frac{1}{p(p-1)} + O\left(\frac{1}{x}\right) \\ &= \log \log x + C_1 + \sum_p \frac{1}{p(p-1)} + O\left(\frac{1}{\log x}\right)\end{aligned}$$

Hence, defining $C_2 = C_1 + \sum_p \frac{1}{p(p-1)}$, we conclude:

$$\sum_{n \leq x} \Omega(n) = x \log \log x + C_2 x + O\left(\frac{x}{\log x}\right)$$

□

1 The Gamma Function

Definition 2. For a complex number z such that $\Re(z) > 0$, we consider the improper integral:

$$\Gamma(z) = \int_0^{+\infty} t^{z-1} e^{-t} dt$$

Theorem 3. $\Gamma(z)$ is absolutely convergent in the half-open plane $\Re(z) > 0$. Furthermore, $\Gamma(z)$ is analytic on $\Re(z) > 0$.

Theorem 4 (Functional Equation). For all $z \in \mathbb{C}$ such that $\Re(z) > 0$, the Gamma function satisfies:

$$\Gamma(z+1) = z\Gamma(z)$$

Proof. We proceed via integration by parts for $\Re(z) > 0$:

$$\Gamma(z+1) = \int_0^{+\infty} t^z e^{-t} dt = \left[-t^z e^{-t}\right]_0^{+\infty} + z \int_0^{+\infty} t^{z-1} e^{-t} dt = 0 + z\Gamma(z)$$

□

Corollary 5. For any positive integer n , $\Gamma(n) = (n-1)!$.

Proof. The proof is by induction.

Base case: $\Gamma(1) = \int_0^{+\infty} e^{-t} dt = 1 = 0!$.

Assume for $k \in \mathbb{N}^*$ that $\Gamma(k) = (k-1)!$. Then:

$$\Gamma(k+1) = k\Gamma(k) = k(k-1)! = k!$$

□

Theorem 6 (Analytic Continuation of $\Gamma(z)$). The Gamma function can be analytically continued to a meromorphic function on $\mathbb{C}$. Furthermore, for every non-negative integer n , $z = -n$ is a simple pole and:

$$\text{Res}(\Gamma, -n) = \frac{(-1)^n}{n!}$$

Proof. For $\Re(z) > 0$, we have $\Gamma(z) = \frac{\Gamma(z+1)}{z}$. The right side of this identity is well-defined for $\Re(z) > -1$ except at $z = 0$. This extends the domain of $\Gamma(z)$ to the region $\Re(z) > -1$. By iterating the functional relation, we can extend $\Gamma(z)$ to $\mathbb{C} \setminus \{-n \mid n \in \mathbb{N}\}$. To find the residue at $z = -n$:

$$\begin{aligned} \text{Res}(\Gamma, -n) &= \lim_{z \rightarrow -n} (z+n)\Gamma(z) = \lim_{z \rightarrow -n} \frac{(z+n)\Gamma(z+n+1)}{z(z+1)(z+2)\dots(z+n)} \\ &= \frac{\Gamma(-n+n+1)}{(-n)(-n-1)\dots(-1)} = \frac{\Gamma(1)}{(-1)^n n!} = \frac{(-1)^n}{n!} \end{aligned}$$

□

Lemma 7. For any real number $0 < a < 1$, we have:

$$\int_{-\infty}^{+\infty} \frac{e^{ax}}{1+e^x} dx = \frac{\pi}{\sin(\pi a)}$$

Proof. Let $f(z) = \frac{e^{az}}{1+e^z}$. We integrate $f(z)$ over a rectangular contour C_R with vertices at $-R, R, R+2\pi i$, and $-R+2\pi i$. Inside this contour, $f(z)$ has a simple pole at $z = \pi i$ with residue $-e^{a\pi i}$. As $R \rightarrow \infty$, the integrals along the vertical segments vanish, yielding the standard evaluation via the Residue Theorem. □

Theorem 8 (Euler's Reflection Formula). For any $z \in \mathbb{C} \setminus \mathbb{Z}$, the Gamma function satisfies:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}$$

Proof. We will show the identity for $0 < \Re(z) < 1$; then by analytic continuation, we extend the identity to $\mathbb{C} \setminus \mathbb{Z}$. Since $0 < \Re(z) < 1$, both integrals converge:

$$\begin{aligned} \Gamma(z)\Gamma(1-z) &= \left(\int_0^\infty t^{z-1} e^{-t} dt \right) \left(\int_0^\infty s^{-z} e^{-s} ds \right) \\ &= \int_0^\infty \int_0^\infty t^{z-1} s^{-z} e^{-(t+s)} dt ds \end{aligned}$$

Let $t = us$, so $dt = sdu$. The double integral becomes:

$$= \int_0^\infty \int_0^\infty (us)^{z-1} s^{-z} e^{-(us+s)} s du ds = \int_0^\infty \int_0^\infty u^{z-1} e^{-s(u+1)} du ds$$

By Fubini's Theorem, integrating with respect to s first:

$$\int_0^\infty u^{z-1} \left(\int_0^\infty e^{-s(u+1)} ds \right) du = \int_0^\infty \frac{u^{z-1}}{u+1} du$$

Using the substitution $u = e^x$, $du = e^x dx$:

$$\Gamma(z)\Gamma(1-z) = \int_{-\infty}^\infty \frac{e^{x(z-1)}}{1+e^x} e^x dx = \int_{-\infty}^\infty \frac{e^{zx}}{1+e^x} dx = \frac{\pi}{\sin(\pi z)}$$

□

Corollary 9. For all $z \in \mathbb{C} \setminus \{0, -1, -2, \dots\}$, $\Gamma(z) \neq 0$.

Theorem 10 (Euler Limit for $\Gamma(z)$). For $\Re(z) > 0$:

$$\Gamma(z) = \lim_{n \rightarrow \infty} \frac{n! n^z}{z(z+1)\dots(z+n)}$$

Proof. Using the Beta function representation:

$$B(z, n+1) = \int_0^1 t^{z-1}(1-t)^n dt = \frac{\Gamma(z)\Gamma(n+1)}{\Gamma(z+n+1)} = \frac{n!\Gamma(z)}{(z+n)(z+n-1)\dots z\Gamma(z)}$$

Thus,

$$\frac{n!n^z}{z(z+1)\dots(z+n)} = n^z \int_0^1 t^{z-1}(1-t)^n dt$$

Making the change of variable $t = \frac{u}{n}$ (so $dt = \frac{du}{n}$), we obtain:

$$n^z \int_0^n \left(\frac{u}{n}\right)^{z-1} \left(1 - \frac{u}{n}\right)^n \frac{du}{n} = \int_0^n u^{z-1} \left(1 - \frac{u}{n}\right)^n du$$

Taking the limit as $n \rightarrow \infty$, since $(1 - \frac{u}{n})^n \rightarrow e^{-u}$ and the integral is dominated by an absolutely convergent integrable function (by DCT):

$$\lim_{n \rightarrow \infty} \frac{n!n^z}{z(z+1)\dots(z+n)} = \int_0^\infty u^{z-1} e^{-u} du = \Gamma(z)$$

□

Theorem 11 (Weierstrass Product). *We have:*

$$\frac{1}{\Gamma(z)} = ze^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}$$

Proof. Using the Euler limit:

$$\frac{1}{z\Gamma(z)} = \lim_{N \rightarrow \infty} \frac{(z+1)\dots(z+N)}{N!N^z} = \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(1 + \frac{z}{n}\right) e^{-z \log N}$$

Inserting $e^{z/n}e^{-z/n} = 1$:

$$= \lim_{N \rightarrow \infty} \prod_{n=1}^N \left(1 + \frac{z}{n}\right) e^{-z/n} \exp\left(z \sum_{n=1}^N \frac{1}{n} - z \log N\right)$$

Since $\sum_{n=1}^N \frac{1}{n} - \log N \rightarrow \gamma$ as $N \rightarrow \infty$, the result follows directly:

$$\frac{1}{z\Gamma(z)} = e^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}$$

□

2 The Riemann Zeta Function

Definition 12. For $\Re(s) > 1$, the Riemann Zeta function is defined as:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

Theorem 13 (Analytic Continuation of $\zeta(s)$). *The Riemann Zeta function can be analytically continued to the half-plane $\Re(s) > 0$ except for a simple pole at $s = 1$ with residue 1.*

Proof. Let $\Re(s) > 1$. Applying the first-order Euler-Maclaurin summation formula to the continuously differentiable function $f(x) = x^{-s}$:

$$\sum_{n=1}^N \frac{1}{n^s} = \int_1^N \frac{dx}{x^s} + \frac{1 + N^{-s}}{2} - s \int_1^N \frac{\tilde{B}_1(x)}{x^{s+1}} dx$$

where $\tilde{B}_1(x) = x - \lfloor x \rfloor - \frac{1}{2}$. Evaluating the first integral:

$$= \frac{N^{1-s} - 1}{1-s} + \frac{1 + N^{-s}}{2} - s \int_1^N \frac{\tilde{B}_1(x)}{x^{s+1}} dx$$

Taking the limit as $N \rightarrow \infty$. Since $\Re(s) > 1$, $N^{1-s} \rightarrow 0$ and $N^{-s} \rightarrow 0$. Thus:

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} - s \int_1^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx$$

Since $\tilde{B}_1(x)$ is uniformly bounded, the integral converges absolutely for $\Re(s) > 0$ and uniformly on any compact subset. Therefore, it defines an analytic function for $\Re(s) > 0$. The rational term $\frac{1}{s-1}$ isolates a simple pole at $s = 1$ with residue 1. $\square$

Corollary 14. For $0 < \Re(s) < 1$, we have:

$$\zeta(s) = -s \int_0^\infty \frac{x - \lfloor x \rfloor}{x^{s+1}} dx$$

Proof. From the previous theorem, for $0 < \Re(s) < 1$:

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} - s \int_1^\infty \frac{x - \lfloor x \rfloor - 1/2}{x^{s+1}} dx$$

We split the integral:

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} - s \int_1^\infty \frac{x - \lfloor x \rfloor}{x^{s+1}} dx + \frac{s}{2} \int_1^\infty \frac{dx}{x^{s+1}}$$

Since $\frac{s}{2} \int_1^\infty x^{-s-1} dx = \frac{1}{2}$, and $\frac{1}{s-1} + 1 = \frac{s}{s-1}$, we have:

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \frac{x - \lfloor x \rfloor}{x^{s+1}} dx$$

Notice that $-s \int_0^1 x^{-s} dx = \frac{-s}{1-s} = \frac{s}{s-1}$. On the interval $(0, 1)$, $\lfloor x \rfloor = 0$, giving:

$$\zeta(s) = -s \int_0^1 \frac{x - \lfloor x \rfloor}{x^{s+1}} dx - s \int_1^\infty \frac{x - \lfloor x \rfloor}{x^{s+1}} dx = -s \int_0^\infty \frac{x - \lfloor x \rfloor}{x^{s+1}} dx$$

$\square$

Theorem 15. For $-1 < \Re(s) < 0$, we have:

$$\zeta(s) = -s \int_0^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx$$

Proof. The idea is to extend the integral $\int_1^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx$ to the domain $\Re(s) > -1$ by applying integration by parts, using the fact that $\frac{d}{dx} \frac{\tilde{B}_2(x)}{2} = \tilde{B}_1(x)$ (where $\tilde{B}_2(x)$ is the second periodic Bernoulli polynomial).

$$\int_1^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx = \left[\frac{\tilde{B}_2(x)}{2x^{s+1}} \right]_1^\infty + \frac{s+1}{2} \int_1^\infty \frac{\tilde{B}_2(x)}{x^{s+2}} dx$$

Since $\tilde{B}_2(x)$ is bounded, this new integral converges absolutely for $\Re(s+2) > 1 \implies \Re(s) > -1$. Now, to verify the exact formula for $-1 < \Re(s) < 0$, evaluate the segment from 0 to 1:

$$\int_0^1 \frac{\tilde{B}_1(x)}{x^{s+1}} dx = \int_0^1 \frac{x - 1/2}{x^{s+1}} dx = \left[\frac{x^{1-s}}{1-s} - \frac{x^{-s}}{-2s} \right]_0^1 = \frac{1}{1-s} + \frac{1}{2s}$$

Multiplying by $-s$ gives $\frac{-s}{1-s} - \frac{1}{2} = \frac{s+1}{2(s-1)}$. But from the previous analytic continuation, $\zeta(s) = \frac{1}{s-1} + \frac{1}{2} - s \int_1^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx$. Notice that $\frac{1}{s-1} + \frac{1}{2} = \frac{s+1}{2(s-1)}$. Thus:

$$\zeta(s) = -s \int_0^1 \frac{\tilde{B}_1(x)}{x^{s+1}} dx - s \int_1^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx = -s \int_0^\infty \frac{\tilde{B}_1(x)}{x^{s+1}} dx$$

□

Theorem 16 (Integral Representation). *For $\Re(s) > 1$:*

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$$

Proof. Recall the definition of the Gamma function:

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$$

For $n \geq 1$, perform the change of variable $t = nx$, yielding $dt = n dx$:

$$\Gamma(s) = \int_0^\infty (nx)^{s-1} e^{-nx} n dx = n^s \int_0^\infty x^{s-1} e^{-nx} dx$$

Dividing by $n^s \Gamma(s)$:

$$\frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} e^{-nx} dx$$

Next, we sum both sides over all $n \geq 1$. We can interchange the sum and the integral by the Monotone/Dominated Convergence Theorem since all terms are positive and the result is absolutely convergent for $\Re(s) > 1$:

$$\zeta(s) = \sum_{n=1}^\infty \frac{1}{n^s} = \frac{1}{\Gamma(s)} \sum_{n=1}^\infty \int_0^\infty x^{s-1} e^{-nx} dx = \frac{1}{\Gamma(s)} \int_0^\infty x^{s-1} \sum_{n=1}^\infty e^{-nx} dx$$

Recognizing the geometric series $\sum_{n=1}^\infty (e^{-x})^n = \frac{e^{-x}}{1-e^{-x}} = \frac{1}{e^x - 1}$, we get:

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$$

□